

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - MEDIUM (Scenario Based Assessment)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 50

ANNEXURE A Scenario Based Assessment

Code of Conduct:

I O. BHANU PRAKASH REDDY - 192372075 *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: obhanuprakashreddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

ANNEXURE

Q1: Student Registration System

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- (a) Enumerate three Collection interfaces suitable for this system.
- (b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- (c) Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

Q2: E-Commerce Cart System

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
- (b) Represent how cart items are stored using a List-based structure.
- (c) Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

Q3: Library Management System

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Q4: Employee Data Processing (10 Marks)

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Q5: Online Voting System (10 Marks)

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting • Display results



SIMATS Online Programming Lab
 JAVA PROGRAMMING

CHIRLI BHANU PRAKASH REDDY
 1027207196

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Student Voting System
10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```

1 import java.util.HashSet;
2 import java.util.HashMap;
3 import java.util.Iterator;
4
5 public class University {
6     public static void main(String[] args) {
7
8         HashSet<String> students = new HashSet<>();
9         HashMap<String, String> courses = new HashMap<>();
10
11         students.add("Ravi");
12         students.add("Rahul");
13     }
14 }
          
```

Input

static input

Output

```

Rahul : Python
Priya : Database
Ravi : Java
          
```

1. code:

```
import java.util.HashSet;
import java.util.HashMap;
import java.util.Iterator;
```

```
public class University {
    public static void main(String[] args) {
```

```
        HashSet<String> students = new HashSet<>();
        HashMap<String, String> courses = new HashMap<>();
```

```
        students.add("Ravi");
        students.add("Rahul");
        students.add("Priya");
```

```
        courses.put("Ravi", "Java");
        courses.put("Rahul", "Python");
        courses.put("Priya", "Database");
```

```
        Iterator<String> it = students.iterator();
```

```
        while (it.hasNext()) {
            String name = it.next();
            System.out.println(name + " : " + courses.get(name));
        }
    }
}
```

2. code:

```
import java.util.ArrayList;
import java.util.Iterator;

public class ShoppingCart {
    public static void main(String[] args) {

        ArrayList<String> cart = new ArrayList<>();

        // Add items
        cart.add("Laptop");
        cart.add("Mouse");
        cart.add("Keyboard");

        // Display items using Iterator
        System.out.println("Shopping Cart Items:");
        Iterator<String> it = cart.iterator();

        while (it.hasNext()) {
            System.out.println(it.next());
        }
    }
}
```

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab title "SIMATS Online Programming Lab", and the user's name "OBILI SHANU PRAKASH REDDY" with a dropdown arrow. The main content area is titled "E-Commerce Cart System" and contains a problem statement and a code editor.

Problem Statement:

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add Items
- Remove an item
- Display all items

Your Code:

```
import java.util.ArrayList;
import java.util.Iterator;

public class ShoppingCart {
    public static void main(String[] args) {

        ArrayList<String> cart = new ArrayList<>();

        cart.add("Laptop");
        cart.add("Mouse");
        cart.add("Keyboard");
    }
}
```

Input:

static input

Output:

```
Shopping Cart Items:
Laptop
Mouse
Keyboard
```

3.code:

```
import java.util.HashMap;

public class Library {
    public static void main(String[] args) {

        HashMap<Integer, String> books = new HashMap<>();

        // Insert book details
        books.put(101, "Java Programming");
        books.put(102, "Data Structures");
        books.put(103, "Database Systems");

        // Retrieve a book
        System.out.println("Book with ISBN 102: " + books.get(102));

        // Display all books
        System.out.println("\nAll Books:");
        for (Integer isbn : books.keySet()) {
            System.out.println(isbn + " : " + books.get(isbn));
        }
    }
}
```

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (CBILU BHANU PRAKASH REDDY). The main content area is titled "Library Management System" and contains a problem statement about a library's book retrieval system. The problem asks for three Map implementations, a schema-like representation, and a Java program using HashMap to insert, retrieve, and display book details. The "Your Code" section shows the Java code for the Library class, which uses HashMap to store book details and prints the book with ISBN 102 and all books. The "Input" section is empty, and the "Output" section shows the expected output: "Book with ISBN 102: Data Structures" and "All Books: 101 : Java Programming, 102 : Data Structures, 103 : Database Systems".

4.code:

```
import java.util.ArrayList;
import java.util.Iterator;
```

```

public class EmployeeList {
    public static void main(String[] args) {

        ArrayList<String> employees = new ArrayList<>();

        employees.add("Ravi - 45000");
        employees.add("Priya - 60000");
        employees.add("Rahul - 75000");

        Iterator<String> it = employees.iterator();

        while (it.hasNext()) {
            String emp = it.next();
            if (emp.contains("60000") || emp.contains("75000")) {
                System.out.println(emp);
            }
        }
    }
}

```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name "SIMATS Online Programming Lab", and the user "CBILL BHANU PRAKASH REDDY". The main section is titled "CO2-AT1-Scenario Based Programs" and displays a list of questions on the left. The selected question, "Q4: Employee Data Processing", is shown in the center. It includes a description of a company processing employee records and three tasks: (a) List three advantages of Generics in Collections, (b) Represent how employee data is stored using List with Generics, and (c) Write a Java program using Iterator to: store employee objects and display employees with salary > 50,000. Below the question, the "Your Code" section shows a Java program that uses ArrayList and Iterator to store and filter employee data. The "Input" section is empty, and the "Output" section shows the filtered results: "Priya - 60000" and "Rahul - 75000".

5.code:

```

import java.util.HashSet;
import java.util.HashMap;

public class VotingSystem {
    public static void main(String[] args) {

        HashSet<String> voters = new HashSet<>();
        HashMap<String, Integer> votes = new HashMap<>();
    }
}

```

```

String voter = "Ravi";
String candidate = "Alice";

if (voters.add(voter)) {
    votes.put(candidate, votes.getOrDefault(candidate, 0) + 1);
    System.out.println("Vote Recorded");
} else {
    System.out.println("Duplicate Vote");
}

System.out.println("Results:");
for (String name : votes.keySet()) {
    System.out.println(name + " : " + votes.get(name));
}
}
}

```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and a user profile for "OBILI SHANU PRAKASH REDDY".

On the left, a "QUESTIONS" sidebar lists five questions (Q1-Q5) with their respective topics and marks. Q5, "Online Voting System - 10 marks", is selected.

The main area displays the question "Online Voting System" with a 10-mark value. The problem statement asks for a Java program that ensures one vote per user and counts votes efficiently. The tasks are:

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting
 - Display results

The "Your Code" section shows a Java program snippet:


```

1 import java.util.*;
2 import java.util.*;
3
4 public class VotingSystem {
5     public static void main(String[] args) {
6
7         HashSet<String> voters = new HashSet<>();
8         HashMap<String, Integer> votes = new HashMap<>();
9
10        String voter = "Ravi";
11        String candidate = "Alice";
12
13        // ... (code continues)

```

The "Input" section has a text area for "stdin input". The "Output" section shows the expected output:


```

Result:
Ravi : 1
Alice : 1

```